

DATS6101 Project 1 Proposal

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Given the ongoing public debate about the necessity and funding of public transportation, specifically the justification for maintaining and expanding urban rail networks, our research will focus on assessing ridership on the New York City subway system.

Our primary research will be led by the question: *“How do NYC subway ridership volumes vary throughout the day during the week of September 15-21, 2024?”*

This broader question will be broken down into more specific, measurable questions such as:

Which hours have the highest ridership average (weekdays vs. weekends)?

Do these hours correspond to commuting periods?

How does subway ridership vary across boroughs within the same time frame?

Exploratory data analysis will provide a clearer picture of what patterns exist within the data, allowing for a more specific question. This research can provide insight into the frequency of use of the NYC subway, informing policymakers and allowing resources to be more appropriately allocated to meet demand. In addition to the primary focus, identifying stations that are outliers for bus-to-subway transfers may allow analysis into where future subway expansion could be most valuable.

The dataset is provided by the New York City government and documents hourly ridership from 2020 to 2024. Given the large number of rows (121 million) within the dataset, our project will generalize average ridership using data from September 15-21, consisting of approximately 553,429 rows of 12 variables.

[Github Link](#)

[Dataset Source](#)